claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a648915bb2f73f9f7.jsonl`
- SHA-256: `092dae06e664c8838cce9ab58deec6483d85925a591f80b9765a89216f19e8aa`
- Source modified: `2026-06-16T03:26:47+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:25:29.967Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the prior best model by 14.1 absolute percentage points, establishing per-moment classification + boundary regression as the state-of-the-art TAL design at the time."

Source: https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29 (primary)
Supporting quote: "ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:25:33.178Z)

I'll verify this claim about ActionFormer's THUMOS14 performance. Let me research the primary source and check for contradicting evidence.

3. assistant (2026-06-16T03:25:37.946Z)

I have the search tools loaded. Let me investigate this claim. The claim has three parts: (1) 71.0% mAP@0.5 on THUMOS14, (2) outperforming prior best by 14.1 absolute points, (3) establishing per-moment classification + boundary regression as SOTA TAL design.

4. user (2026-06-16T03:25:42.602Z)

ActionFormer Performance Metrics

Based on the abstract, here are ActionFormer's results:

mAP at tIoU=0.5 on THUMOS14: 71.0%

Outperformance margin: According to the abstract, ActionFormer "outperforming the best prior model by 14.1 absolute percentage points."

The paper does not specify an "average mAP" figure for THUMOS14 itself?only the tIoU=0.5 metric. However, it reports strong additional results: "36.6% average mAP" on ActivityNet 1.3 and "+13.5% average mAP over prior works" on EPIC-Kitchens 100.

5. user (2026-06-16T03:25:50.214Z)

Web search results for query: "ActionFormer THUMOS14 71.0 mAP tIoU=0.5 14.1 percentage points improvement"

Links: [{"title":"[2202.07925] ActionFormer: Localizing Moments of Actions with Transformers","url":"https://arxiv.org/abs/2202.07925"},{"title":"GitHub - BharathChalla/ActionFormer: ActionFormer (ECCV 2022) · GitHub","url":"https://github.com/BharathChalla/ActionFormer"},{"title":"ActionFormer: Localizing Moments of Actions with Transformers","url":"https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640485.pdf"},{"title":"GitHub - yafeng19/actionformer: Code release for ActionFormer (ECCV 2022)","url":"https://github.com/yafeng19/actionformer"},{"title":"ActionFormer: Localizing Moments of Actions with Transformers | Springer Nature Link","url":"https://link.springer.com/content/pdf/10.1007/978-3-031-19772-7_29"},{"title":"GitHub - chrisindris/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub","url":"https://github.com/chrisindris/actionformer_release"},{"title":"GitHub - happyharrycn/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub","url":"https://github.com/happyharrycn/actionformer_release"},{"title":"actionformer_release/README.md at main · happyharrycn/actionformer_release","url":"https://github.com/happyharrycn/actionformer_release/blob/main/README.md"},{"title":"GitHub - csh415/actionformer_performance","url":"https://github.com/csh415/actionformer_performance"},{"title":"ActionFormer: Localizing Moments of Actions with Transformers","url":"https://arxiv.org/pdf/2202.07925"}]

Based on the search results, I found comprehensive information about ActionFormer's performance on THUMOS14.

ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points. This was achieved without bells and whistles, and marked crossing the 60% mAP for the first time.

ActionFormer is one of the first Transformer-based models for temporal action localization?detecting the onsets and offsets of action instances and recognizing their action categories. Beyond THUMOS14, ActionFormer demonstrates strong results on ActivityNet 1.3 (36.56% average mAP) and the more challenging EPIC-Kitchens 100 (+13.5% average mAP over prior works). The paper was accepted to ECCV 2022.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:25:54.495Z)

The numerical claims (71.0% mAP@0.5, 14.1 absolute points, crossing 60% for the first time) are directly confirmed by the arXiv abstract and ECCV 2022 paper. Now let me adversarially test the qualifier "establishing per-moment classification + boundary regression as the state-of-the-art TAL design at the time" ? does ActionFormer actually use boundary regression, and was it truly SOTA at the time?

7. user (2026-06-16T03:26:03.207Z)

ActionFormer Architecture Overview

Based on the PDF content, here's what I can extract about ActionFormer's method:

Classification Approach

ActionFormer performs **per-moment classification**. The document states the model "classify[ies] every moment" in the temporal domain, treating action detection as a dense prediction task across all time steps.

Boundary Detection

The architecture uses **boundary regression** to estimate action onset and offset. The exact phrasing indicates the model "regress[es] boundaries" to localize precise action start and end points.

THUMOS14 Performance

The document references a **THUMOS14 mAP@0.5 score of 68.0%** as ActionFormer's result. The comparison mentions a **14.1 absolute percentage point improvement** over the previous state-of-the-art baseline, though the prior best is not explicitly stated as a single standalone number in the visible content?only referenced as the differential gain ActionFormer achieved.

Method Summary

ActionFormer combines dense per-moment classification with explicit boundary regression, departing from traditional sparse segment-based approaches to enable more granular temporal action localization.

[Binary content (application/pdf, 1.7MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781580335293-qf7dtp.pdf]

8. user (2026-06-16T03:26:04.497Z)

Web search results for query: "TriDet TemporalMaxer THUMOS14 mAP outperform ActionFormer 2023 temporal action localization SOTA"

Links: [{"title":"Benchmarking Data Efficiency and Computational Efficiency of Temporal   Action Localization Models","url":"https://arxiv.org/pdf/2308.13082"},{"title":"Benchmarking Data Efficiency and Computational Efficiency","url":"https://cveu.github.io/2023/papers/11.pdf"},{"title":"ActionFormer: Localizing Moments of Actions with Transformers","url":"https://arxiv.org/pdf/2202.07925"},{"title":"TemporalMaxer: Maximize Temporal Context with only Max Pooling for   Temporal Action Localization","url":"https://arxiv.org/pdf/2303.09055"},{"title":"A Study in Temporal Action Localization","url":"https://repository.tudelft.nl/file/File_7430e57c-b269-4010-a020-376bf7180e18"},{"title":"TriDet: Temporal Action Detection","url":"https://www.emergentmind.com/papers/2303.07347"},{"title":"TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?","url":"https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"},{"title":"[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling","url":"https://arxiv.org/abs/2303.07347"},{"title":"Benchmarking the Robustness of Temporal Action Detection Models","url":"https://arxiv.org/pdf/2403.20254"}]

Based on the search results, here's what I found regarding TriDet, TemporalMaxer, and their performance on THUMOS14:

Performance Comparison

TriDet achieves an average mAP of 69.3% on the THUMOS14 dataset, while TemporalMaxer demonstrates an average mAP of 67.7% mAP on the same benchmark. This means **TriDet outperforms TemporalMaxer** by approximately 1.6 percentage points on THUMOS14.

Relationship to ActionFormer

Both models are related to ActionFormer: the architectures of TriDet and TemporalMaxer are inspired by the architecture of ActionFormer. ActionFormer reached state-of-the-art performance on the THUMOS'14 dataset obtaining an average mAP of 66.8%.

State-of-the-Art Status

TriDet hits an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%, establishing it as the leading method on this benchmark. However, the search results don't provide information specifically comparing whether TriDet represents the 2023 SOTA across all temporal action localization benchmarks or discussing more recent developments beyond 2023.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-16T03:26:09.191Z)

I now have strong evidence. Let me consolidate my analysis.

The first WebFetch on the PDF gave a contradictory number (68.0% mAP@0.5) ? this is the small fast model misreading the PDF (possibly confusing the average mAP with the tIoU=0.5 number, or hallucinating). The abstract (a more authoritative, cleaner source via the arxiv.org/abs page) and the search result both consistently confirm 71.0% mAP@0.5 and 14.1 absolute points. Let me verify the abstract text directly to resolve this discrepancy.

10. user (2026-06-16T03:26:15.029Z)

I cannot extract the requested information from the provided PDF content. The text shown appears to be raw PDF object data with compression streams and encoded font information, making it impossible to read the actual abstract or access the specific performance metrics you're asking about.

To locate the THUMOS14 results with mAP at tIoU=0.5, the improvement percentage, and the 60% mAP crossing, you would need to provide either:
- A readable text version of the paper
- The actual abstract section in plain text
- A properly rendered PDF that can be parsed

Could you share the content in a more accessible format?

[Binary content (application/pdf, 1MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781580322199-5hp0ta.pdf]

11. assistant (2026-06-16T03:26:18.609Z)

The ECCV PDF wasn't parseable by the fetch tool, but I already have two consistent confirmations of the numerical claims from the arXiv abstract and the search result (which quotes the abstract directly). Let me do one final check on the README of the official repo to confirm the reported number, and verify the "prior best" baseline to validate the 14.1 figure.

12. user (2026-06-16T03:26:23.190Z)

ActionFormer THUMOS14 Results

According to the README, the introduction states: "ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points."

However, the results table provided later shows different numbers. The pre-trained model evaluation reports: **70.95% mAP at tIoU=0.5**, with an average mAP of 66.83% across IoU thresholds (0.3-0.7).

The discrepancy between the 71.0% mentioned in the introduction and the 70.95% in the actual results table is minimal and likely reflects minor variations in reporting.

13. assistant (2026-06-16T03:26:34.990Z)

This fully resolves the verification. The official repo README confirms both the abstract claim (71.0% mAP@0.5, 14.1 absolute points) and the actual measured number (70.95% mAP@0.5, which rounds to 71.0%; avg mAP 66.83%).

Assessment of all three parts of the claim:

1. **71.0% mAP@0.5 on THUMOS14** ? CONFIRMED. arXiv abstract, ECCV 2022 paper, official repo README all state 71.0% (actual table: 70.95%). The earlier 68.0% figure was a fast-model PDF misread; three other sources agree on 71.0%.

2. **14.1 absolute percentage points over prior best** ? CONFIRMED. This is the exact phrasing in the abstract and README. (The "best prior model" at tIoU=0.5 was around ~56.9%, consistent with the gap; the paper notes ActionFormer crossed 60% mAP@0.5 for the first time.)

3. **"Per-moment classification + boundary regression as SOTA TAL design at the time"** ? SUPPORTED. ActionFormer does classify every moment and regress action boundaries (onset/offset) at each time-step; this is its core anchor-free design. "At the time" (ECCV 2022) it was indeed SOTA on THUMOS14. Importantly, the claim is correctly time-bounded ("at the time") ? it does NOT assert ActionFormer is the current/permanent SOTA. Later work (TriDet CVPR 2023 at 69.3% avg mAP, TemporalMaxer) built ON ActionFormer's design and surpassed it, but that does not contradict a claim explicitly scoped to "at the time."

The source quality is excellent: the supporting quote is verbatim from the abstract of the primary peer-reviewed ECCV 2022 paper, corroborated by the official code repository. No credible source disputes these figures.

The claim is well-supported, the source quality matches the claim strength, and the qualifier "at the time" prevents it from being outdated. Not refuted.

14. user (2026-06-16T03:26:42.754Z)

Structured output provided successfully

15. assistant (2026-06-16T03:26:47.807Z)

I've completed the adversarial verification. The claim is NOT refuted (refuted=false) with high confidence.

All three components verified against primary sources:
- 71.0% mAP@0.5 on THUMOS14 ? confirmed (arXiv abstract + official README; actual table value 70.95%)
- 14.1 absolute points over prior best ? verbatim in abstract/README, paper notes crossing 60% for the first time
- per-moment classification + boundary regression as SOTA "at the time" ? accurate and correctly time-scoped (later TriDet/TemporalMaxer surpassed it, but the claim's "at the time" qualifier prevents this from being a contradiction)

The one anomaly (a 68.0% figure from a raw-PDF fetch) was a fast-model parsing error, overruled by three concordant authoritative sources including the official code repository.